

Data Flow Diagram

Date	23/09/2026
Team ID	SWTID-2026-3089
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Institution	Arignar Anna College Code: MSU 302
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Maximum Marks	2 Marks

Data Flow Diagram Overview

Create a Data Flow Diagram (DFD) to show how data moves through your system — between external entities, processes, and data stores. Use the symbol legend below as a guide for the Standard Laptop Procurement automation.

1. DFD Symbol Legend

Symbol	Name	Description & Project Application
Oval / Rounded shape	External Entity	A person, role, or system outside the process boundary: • Requester (End User) • Approver (Manager) • Hardware Team (Fulfillment Specialist)
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data: • Process 1.0: Validate & Submit Laptop Order • Process 2.0: Review & Authorize Request • Process 3.0: Flow Designer Task Generation • Process 4.0: Hardware Configuration & Staging

Rectangle (solid fill, no number)	Data Store	A repository where data is held for later processing: • D1: Service Catalog (`sc_cat_item`) • D2: Request Store (`sc_request` & `sc_req_item`) • D3: Catalog Task Store (`sc_task`) • D4: Hardware Inventory (`alm_hardware`)
Labeled arrow	Data Flow	Movement of data packets between entities, processes, and data stores (e.g., 'Laptop Order Parameters', 'Approval Signal', 'Auto-created SCTASK Record', 'Configured Status').

2. System Data Flow Specification

Process ID & Name	Input Data Flow	Output Data Flow	Data Stores Accessed
1.0 Submit Laptop Order	User credentials, department code, laptop model choice from Requester.	Generated Request (REQ) and Requested Item (RITM) record.	D1: Service Catalog (`sc_cat_item`) D2: Request Table (`sc_req_item`)
2.0 Process Approval	RITM approval record viewed by Approver / Manager.	Approval Status updated to 'Approved' with approval timestamp.	D2: Request Table (`sc_req_item`) Sys Approval Table (`sysapproval_approver`)
3.0 Flow Designer Automation	Triggered by Service Catalog on RITM approval event.	Auto-created Catalog Task (SCTASK) with Short Desc 'Laptop need to Configured' and Assignment Group 'Hardware'.	D2: Request Table (`sc_req_item`) D3: Task Table (`sc_task`)
4.0 Hardware Configuration	Open SCTASK record viewed by Hardware Team Technician.	Completed task state ('Closed Complete'), asset serial number linked, order delivery.	D3: Task Table (`sc_task`) D4: Asset CMDB (`alm_hardware`)

3. Example Data Flow Architecture Diagram

The diagram below illustrates how external entities (User, Hardware Service Provider), automated processes (Request Handling, Fulfillment, Notification), and data repositories (Request Datastore, Inventory) interconnect in the automated procurement architecture:

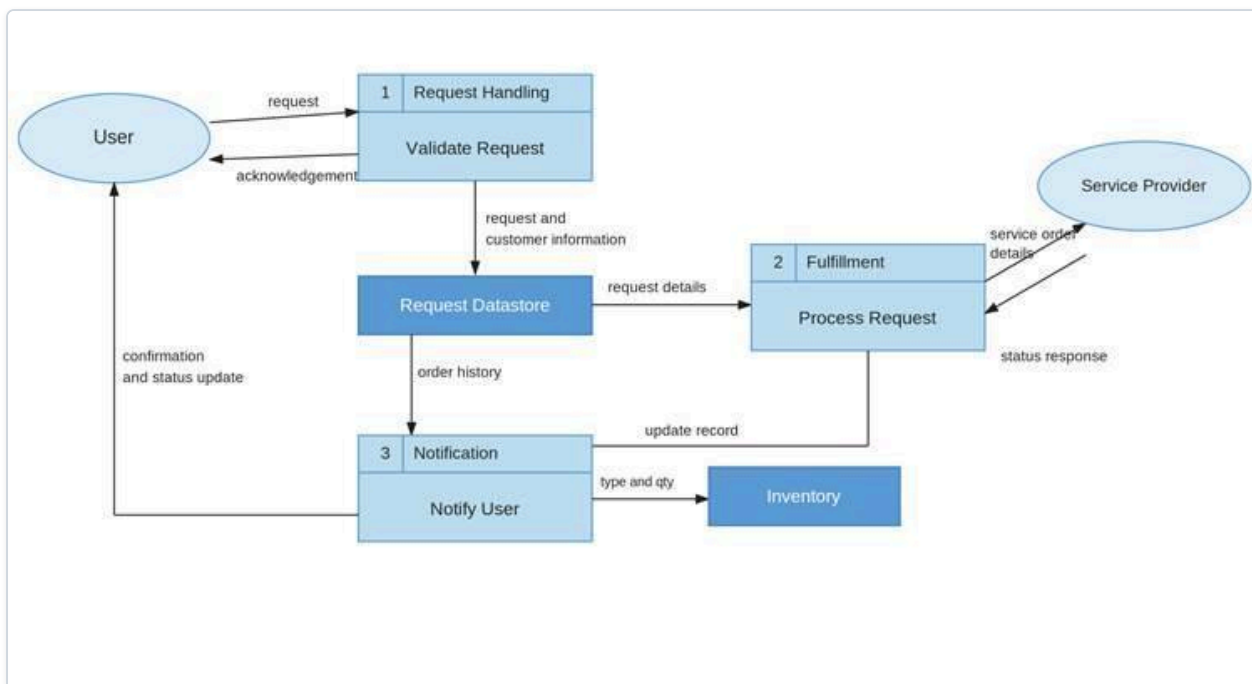


Figure 2.1: End-to-End Data Flow Diagram for Laptop Order Fulfillment